University of Waterloo

# ACTSC 446

Mathematics of Financial Markets

Lecture Notes

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**Standing conventions**

Unless stated otherwise, $0 \le t \le T$, $S_u \ge 0$, $K \ge 0$, and $r$ is a constant continuously compounded risk-free rate. When dividends are continuous, $\delta \ge 0$ denotes a constant proportional dividend yield. The stock price is *ex-dividend*. We explicitly impose $r \ge 0$ whenever a proof needs it. Discounted European bounds and forward formulas themselves allow negative $r$.

**What is expanded or corrected**

The bank account $B_u = e^{ru}$ is distinguished from a zero-coupon bond paying one dollar at maturity. Arbitrage includes nonnegative payoffs with a strictly positive probability of gain, and strategies must be self-financing. Discrete-dividend forward formulas require dividends known at the valuation time. Forward delivery prices are distinguished from contract values. American exercise arguments cover the short writer's obligations at every permitted exercise time. Payoff, premium, and financed profit are kept separate. The final sections clarify the lecture's butterfly convention and the limitations of claims about volatility and risk-adjusted performance.

**Notation**

| Symbol | Meaning |
|---|---|
| $S_u$ | Ex-dividend stock price at time $u$ |
| $B_u = e^{ru}$ | Bank account starting at one dollar at time zero |
| $D(t,T) = e^{-r(T-t)}$ | Price at $t$ of one dollar delivered at $T$ |
| $A(t,T)$ | Price at $t$ of a prepaid claim to one share at $T$ |
| $F(t,T)$ | Delivery price of a new zero-value forward |
| $f_t(K,T)$ | Value at $t$ of a long forward with fixed delivery price $K$ |
| $c(K,t,T),\ p(K,t,T)$ | European call and put prices |
| $C(K,t,T),\ P(K,t,T)$ | American call and put prices |
| $x^+ = \max\{x, 0\}$ | Positive part of a real number |

# Chapter 1

# Introduction to Derivatives Markets

## 1.1 Financial markets and trading conventions

### 1.1.1 Assets, time, and uncertainty

**Definition 1.1** (Asset and price process)**.** An asset is a financial instrument that can be bought or sold. Its price process $(S_u)_{u\geq 0}$ specifies its market value through time. At a future time, $S_u$ may be random. At the valuation time $t$, the observed $S_t$ is known to the trader.

One may formalize uncertainty on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_u)_{u\geq 0}, \mathbb{P})$, with $S_u$ measurable with respect to $\mathcal{F}_u$. This says that the price at $u$ is observable using information available by $u$. Part I does not specify the law of $(S_u)$; most arguments below hold path by path. At a later valuation time $t$, probabilistic statements are understood relative to the information then available. In particular, a claim of strict price dominance requires a positive conditional probability of strict payoff dominance at that valuation state. At time zero this reduces to the usual unconditional statement.

**Definition 1.2** (Model-free argument)**.** A pricing argument is model-free here if it relies on trading assumptions and payoff identities or inequalities, without specifying a probability distribution or stochastic differential equation for the underlying.

Model-free does not mean assumption-free. Dividend rules, financing opportunities, short-sale access, settlement conventions, and exercise rights all matter.

**Definition 1.3** (Perfect capital market conventions)**.** For the arguments in these notes, assume:

(i) no transaction costs and a common buying and selling price;

(ii) arbitrary fractional positions and unrestricted short positions;

(iii) execution at observed prices, with no delay or price impact;

(iv) borrowing and lending at the same specified risk-free rate;

(v) no counterparty default on promised contractual cash flows;

(vi) no trading advantage arising from different information sets.

Continuous-time exercise arguments additionally require the ability to transact when exercise occurs.

*Remark* 1.1 (Credit risk versus price risk). Removing counterparty default risk does not make a stock risk-free. Its market value may still fluctuate or become zero. A stock price loss is different from failure to honour a promised derivative payment.

### 1.1.2 The bank account and zero-coupon bonds

**Definition 1.4** (Bank account)**.** The risk-free bank account with $B_0 = 1$ is

$$B_u = e^{ru}.$$

An amount $a$ invested at time $t$ has value $aB_u/B_t = ae^{r(u-t)}$ at $u \geq t$. A negative amount represents borrowing.

**Definition 1.5** (Zero-coupon bond)**.** A zero-coupon bond with maturity $T$ and face value one pays exactly one dollar at $T$, with no intermediate payments. Its time-$t$ price is denoted by $D(t, T)$.

**Proposition 1.1** (Discounting a deterministic payment)**.** *In an arbitrage-free market with the stated bank account,*

$$D(t, T) = \frac{B_t}{B_T} = e^{-r(T-t)}.$$

*A deterministic payment $a$ at $T$ therefore has value $ae^{-r(T-t)}$ at $t$.*

*Proof.* Purchase $1/B_T$ units of the bank account. The cost and terminal value are

$$\text{cost at } t = \frac{B_t}{B_T} = e^{-r(T-t)},$$
$$\text{value at } T = \frac{B_T}{B_T} = 1.$$

A bond with the same cash flows must have the same price by the law of one price, proved in Section 1.2. Multiplying the position by $a$ gives the general formula. □

*Remark* 1.2 (Two different assets). The lecture writes $B_t = e^{rt}$ and calls this a risk-free bond. We use the more precise name *bank account*. A bond that pays one at $T$ has price $e^{-r(T-t)}$, whereas one dollar placed in the bank account at zero becomes $e^{rT}$ at $T$.

*Remark* 1.3 (The sign of the interest rate). For $r > 0$, a future dollar costs less than a current dollar. For $r = 0$, the two have equal value. For $r < 0$, the future dollar costs more. None of these possibilities contradicts no-arbitrage by itself.

### 1.1.3 Portfolios, cash flows, and self-financing

**Definition 1.6** (Portfolio value)**.** For assets with prices $S_u^1, \ldots, S_u^m$ and holdings $\theta_u^1, \ldots, \theta_u^m$, the marked value is

$$V_u = \sum_{i=1}^{m} \theta_u^i S_u^i.$$

Positive holdings are long positions; negative holdings are short positions.

For example, three shares and 200 units of the bank account have value $3S_u + 200B_u$. Three shares financed by borrowing 200 dollars at time zero have value $3S_u - 200B_u$. At a later starting time $t$, a loan of 200 dollars instead contributes $-200B_u/B_t$.

**Definition 1.7** (Self-financing strategy)**.** A strategy is self-financing if, after its initial setup, changes in holdings and contractual obligations are funded entirely by asset sales, borrowing, and cash flows from its existing positions. No outside money is injected or withdrawn without being recorded.

At a rebalancing time $u$ with no simultaneous dividend, changing holdings from $(\theta, \beta)$ to $(\theta', \beta')$ requires

$$\begin{aligned}(\theta' - \theta)S_u + (\beta' - \beta)B_u &= 0,\\ \theta' S_u + \beta' B_u &= \theta S_u + \beta B_u.\end{aligned}$$

Thus rebalancing redistributes wealth; it does not create wealth. If dividends are received, they can finance additional shares or be put in the bank account.

*Remark* 1.4 (Value versus cash flow). Buying an asset priced at $v$ produces an initial cash flow of $-v$ but acquires value $v$. Selling it produces cash flow $+v$ and a position with value $-v$. Cash-flow tables below use *positive for cash received*. A portfolio's initial cost has the opposite sign to its setup cash flow.

*Remark* 1.5 (Short sales and dividends). A short seller owes any distributions paid to the owner of the borrowed stock. A short share contributes $-S_T$ upon closing at $T$, but this is not its entire cash-flow history if dividends occur before $T$.

### 1.1.4 Derivatives as contractual claims

**Definition 1.8** (Derivative)**.** A derivative is a contract whose payments are determined by specified underlying market quantities. A European terminal claim may be written as

$$H = h(S_T),$$

while a path-dependent claim may depend on the whole observed path $(S_u)_{0 \le u \le T}$.

Underlying quantities may include stocks, bonds, currencies, indices, commodities, or rates. Derivatives can transfer risk, create compensation incentives, or take speculative exposures. A small initial premium need not mean a small exposure: contractual payments can be much larger than the setup cost.

## 1.2 No-arbitrage and the comparison principle

*Source coverage: lecture slides 8–12..*

### 1.2.1 A precise definition of arbitrage

**Definition 1.9** (Arbitrage)**.** Starting at time $t$, an arbitrage is an admissible self-financing strategy with zero initial wealth, terminal wealth $Z_T \geq 0$ almost surely, and

$$\mathbb{P}(Z_T > 0) > 0.$$

Admissibility rules out strategies such as unlimited doubling schemes. All explicit constructions here use finite positions and financed contractual obligations.

A strategy with negative initial cost and nonnegative future cash flows also generates arbitrage: invest the initial receipt. If its cost is $-a < 0$, the additional bank holding pays $ae^{r(T-t)} > 0$ at $T$.

*Remark* 1.6 (Relation to the lecture definition)*.* The slides use $V_0 \leq 0$ and $V_T > 0$. Guaranteed positive terminal wealth is a particularly transparent arbitrage. The standard probabilistic definition also includes a payoff that is positive only on an event of positive probability and never negative. Our explicit mispricing constructions usually produce a deterministic positive bank balance in every state.

### 1.2.2 The law of one price

**Theorem 1.1** (Law of one price)**.** *Suppose two attainable strategies $X$ and $Y$ have identical remaining cash flows, including identical terminal liquidation values, and no unaccounted intermediate payments. Then their prices at time $t$ are equal.*

*Proof.* Suppose $X_t > Y_t$ and let $a = X_t - Y_t > 0$. Buy $Y$, sell $X$, and invest $a$ in the bank account. The resulting strategy has value

$$Z_u = Y_u - X_u + a\frac{B_u}{B_t}.$$

At setup,

$$\begin{aligned} Z_t &= Y_t - X_t + (X_t - Y_t) \\ &= 0. \end{aligned}$$

Because the positions have matching cash flows, they cancel, leaving

$$\begin{aligned} Z_T &= Y_T - X_T + a\frac{B_T}{B_t} \\ &= ae^{r(T-t)} > 0. \end{aligned}$$

This is arbitrage. If $Y_t > X_t$, reverse the positions. Therefore $X_t = Y_t$. □

*Remark* 1.7 (Why terminal values alone may not suffice)*.* Two assets with equal ex-dividend values at $T$ may have paid different dividends before $T$. Their terminal prices alone do not identify their total cash flows. Either match all intermediate payments or explicitly reinvest them and compare terminal wealth.

### 1.2.3 Payoff dominance and strict inequalities

**Proposition 1.2** (Monotonicity of attainable prices)**.** *Let $X$ and $Y$ be terminal-payoff claims with no intermediate payments. If $X_T \geq Y_T$ almost surely, then $X_t \geq Y_t$. If additionally $\mathbb{P}(X_T > Y_T) > 0$, then $X_t > Y_t$ under the stated arbitrage definition.*

*Proof.* If $X_t < Y_t$, buy $X$, sell $Y$, and invest the initial receipt $Y_t - X_t$. Then

$$\begin{aligned} Z_t &= X_t - Y_t + (Y_t - X_t) = 0, \\ Z_T &= X_T - Y_T + (Y_t - X_t)e^{r(T-t)} > 0. \end{aligned}$$

Thus $X_t \geq Y_t$. If the terminal inequality is strict with positive probability but $X_t = Y_t$, the zero-cost position $X - Y$ already has nonnegative payoff and positive probability of gain. This is also arbitrage, proving strictness. □

**Corollary 1.1** (Nonnegative payoff and deterministic bounds)**.** *If $H \geq 0$, its price is nonnegative. More generally, if deterministic constants satisfy $a \leq H \leq b$, then*

$$aD(t,T) \leq \text{price}_t(H) \leq bD(t,T).$$

*Proof.* Compare $H$ with zero, or with bonds paying $a$ and $b$ at $T$. □

*Remark* 1.8 (Weak versus strict wording). A payoff that is merely "higher" in the sense $X_T \geq Y_T$ supports a weak price inequality. Strictness needs a genuine strictly positive gain on an event that is not null. This distinction matters when payoffs agree on every reachable stock-price state.

### 1.2.4 Uniqueness of the risk-free rate

**Proposition 1.3** (A common risk-free rate)**.** *Two freely accessible risk-free accounts, with identical currency, maturity, and borrowing/lending permissions, cannot offer different continuously compounded rates.*

*Proof.* Suppose $r_2 > r_1$. Borrow one dollar at $r_1$ and lend it at $r_2$. The initial cash flow is $1 - 1 = 0$, and the terminal net wealth is

$$\begin{aligned} Z_T &= e^{r_2(T-t)} - e^{r_1(T-t)} \\ &= e^{r_1(T-t)} \left( e^{(r_2 - r_1)(T-t)} - 1 \right) > 0. \end{aligned}$$

This contradicts no-arbitrage. Interchanging the accounts rules out $r_1 > r_2$. □

*Remark* 1.9. Different borrowing and lending rates in an imperfect market do not automatically imply arbitrage. The proof requires borrowing at the lower rate and lending at the higher rate in unrestricted amounts.

### 1.2.5 A reusable proof method

To prove a price identity, first match cash flows. To prove a bound, construct a portfolio that covers the claim in every state. If the claimed price relation fails, buy the cheaper cash-flow stream, sell the more expensive one, and invest the difference. For American options, "every state" must include *every permitted exercise time of the short position.*

## 1.3 Forwards, dividends, and futures

*Source coverage: lecture slides 14–40..*

### 1.3.1 The forward contract and its two prices

**Definition 1.10** (Forward contract)**.** A forward with delivery date $T$ and delivery price $K$ obliges its long holder to pay $K$ and receive one unit of the underlying at $T$. The short holder must deliver that unit and receives $K$.

The net values exchanged at maturity are

$$\begin{aligned} H_T^{\text{long}} &= S_T - K, \\ H_T^{\text{short}} &= K - S_T = -H_T^{\text{long}}. \end{aligned}$$

Physical delivery and cash settlement have the same net value in the frictionless setting.

**Definition 1.11** (Forward price and forward value)**.** The *forward price* $F(t,T)$ is the delivery price that makes a new forward have zero value at time $t$. The *forward value* $f_t(K,T)$ is the market value at $t$ of an existing contract whose delivery price is the fixed number $K$.

Thus $f_t(F(t,T),T) = 0$. The number $K$ does not reset when market conditions change; the value of the existing contract does.

### 1.3.2 No-dividend forward replication

**Theorem 1.2** (Forward value without dividends)**.** *If the stock pays no dividends on* $(t,T]$*, then*

$$\begin{aligned} f_t(K,T) &= S_t - KD(t,T), \\ F(t,T) &= S_t e^{r(T-t)}. \end{aligned}$$

*Proof.* Buy one share and borrow $KD(t,T)$. The loan grows to $K$ at maturity. The portfolio has cost $S_t - KD(t,T)$ and payoff

$$\begin{aligned} V_T &= S_T - KD(t,T)e^{r(T-t)} \\ &= S_T - K. \end{aligned}$$

This replicates the long forward, so the law of one price gives the first formula. A new forward has value zero, hence

$$\begin{aligned} 0 &= S_t - F(t,T)e^{-r(T-t)}, \\ F(t,T)e^{-r(T-t)} &= S_t, \\ F(t,T) &= S_t e^{r(T-t)}. \end{aligned}$$

□

| Position | Cash flow at $t$ | Cash flow at $T$ |
|---|---|---|
| Buy one share | $-S_t$ | $S_T$ |
| Borrow $KD(t,T)$ | $+KD(t,T)$ | $-K$ |
| Total | $-S_t + KD(t,T)$ | $S_T - K$ |

**Arbitrage when the quoted delivery price is too high**

Suppose a newly offered zero-cost forward has delivery price $K > S_t e^{r(T-t)}$. Buy one share, borrow $S_t$, and short the forward. Then

$$\begin{aligned}\text{initial cash flow} &= -S_t + S_t + 0 = 0,\\ \text{terminal wealth} &= S_T - S_t e^{r(T-t)} + (K - S_T)\\ &= K - S_t e^{r(T-t)} > 0.\end{aligned}$$

**Arbitrage when the quoted delivery price is too low**

If $K < S_t e^{r(T-t)}$, short one share, invest $S_t$, and go long the forward. Then

$$\begin{aligned}\text{initial cash flow} &= S_t - S_t + 0 = 0,\\ \text{terminal wealth} &= -S_T + S_t e^{r(T-t)} + (S_T - K)\\ &= S_t e^{r(T-t)} - K > 0.\end{aligned}$$

Both directions are necessary: one construction gives one inequality, and its reverse gives the other.

**Example 1.1** (A contract need not remain worth zero)**.** Let $S_0 = 100$, $r = 0.05$, and $T = 1$. The original delivery price is $K = 100e^{0.05} \approx 105.1271$. At $t = 0.5$, suppose $S_t = 110$. Then

$$\begin{aligned}f_{0.5}(K, 1) &= 110 - 105.1271e^{-0.025}\\ &\approx 7.4685.\end{aligned}$$

The original long forward is now valuable. Its zero initial cost was a statement about inception, not all later times.

### 1.3.3 Continuous proportional dividends

**Definition 1.12** (Continuous dividend yield)**.** A stock with constant dividend yield $\delta$ pays $\delta S_u \,\mathrm{d}u$ per share during a short interval of length $\mathrm{d}u$. The quoted $S_u$ excludes cash already paid as dividends.

**Lemma 1.1** (Reinvestment grows the share count)**.** *If dividends are immediately reinvested, a position of $n_t$ shares grows to*

$$n_u = n_t e^{\delta(u-t)}.$$

*In particular, buying $e^{-\delta(T-t)}$ shares at $t$ and reinvesting dividends produces exactly one share at $T$.*

*Proof.* The dividend received on $n_u$ shares is $n_u \delta S_u \,\mathrm{d}u$. Reinvesting it buys

$$\begin{aligned}\mathrm{d}n_u &= \frac{n_u \delta S_u \,\mathrm{d}u}{S_u}\\ &= \delta n_u \,\mathrm{d}u.\end{aligned}$$

For a positive share price, solving this ordinary differential equation gives

$$\begin{aligned}\frac{\mathrm{d}n_u}{n_u} &= \delta\,\mathrm{d}u,\\ \log n_u - \log n_t &= \delta(u-t),\\ n_u &= n_t e^{\delta(u-t)}.\end{aligned}$$

Choose $n_t = e^{-\delta(T-t)}$ to obtain $n_T = 1$. The continuous-yield model convention extends this construction through zero-price states; there is then no positive dividend cash flow to finance. Equivalently, the reinvested position's wealth and dividend-financed share rule specify the strategy directly. □

**Definition 1.13** (Prepaid forward)**.** A prepaid forward is a claim paid for at $t$ that delivers one share at $T$, with no delivery payment at $T$ and no intervening dividends to its holder. Denote its price by $A(t,T)$.

The lemma and replication give

$$A(t,T) = S_t e^{-\delta(T-t)}.$$

This is not generally $S_t$: the current shareholder has dividend rights that the prepaid-forward holder does not receive.

**Theorem 1.3** (Forward pricing with continuous dividends)**.** *Under the continuous-yield convention,*

$$\begin{aligned}f_t(K,T) &= S_t e^{-\delta(T-t)} - Ke^{-r(T-t)},\\ F(t,T) &= S_t e^{(r-\delta)(T-t)}.\end{aligned}$$

*Proof.* Buy $e^{-\delta(T-t)}$ shares and reinvest dividends until $T$. Borrow $Ke^{-r(T-t)}$. The stock holding becomes one share and the loan becomes $K$, so the terminal payoff is $S_T - K$. Consequently,

$$\begin{aligned}f_t(K,T) &= A(t,T) - KD(t,T)\\ &= S_t e^{-\delta(T-t)} - Ke^{-r(T-t)}.\end{aligned}$$

Set $K = F(t,T)$ and solve $f_t(K,T) = 0$:

$$\begin{aligned}F(t,T)e^{-r(T-t)} &= S_t e^{-\delta(T-t)},\\ F(t,T) &= S_t e^{r(T-t)-\delta(T-t)}\\ &= S_t e^{(r-\delta)(T-t)}.\end{aligned}$$

□

*Remark* 1.10 (Why dividends reduce the delivery price). Holding a share delivers dividends; holding a forward does not. Financing the share costs interest, while its dividends offset part of that cost. The net carry rate is $r - \delta$. The forward price is below spot when $\delta > r$, equal to spot when $\delta = r$, and above spot when $\delta < r$.

**Example 1.2.** For $S_t = 100$, $T - t = 0.5$, $r = 0.04$, and $\delta = 0.015$,

$$\begin{aligned}A(t,T) &= 100e^{-0.015(0.5)} \approx 99.2528,\\ F(t,T) &= 100e^{(0.04-0.015)(0.5)} \approx 101.2578.\end{aligned}$$

The prepaid price and the delivery price refer to different payment dates.

### 1.3.4 Known discrete cash dividends

**Theorem 1.4** (Forward pricing with deterministic cash dividends)**.** *Suppose the stock pays known cash amounts $d_i$ at dates $t < t_1 < \cdots < t_m \leq T$. Assume the forward holder does not receive these dividends and that a dividend at $T$ is paid before delivery. Write*

$$I_t = \sum_{i=1}^{m} d_i e^{-r(t_i - t)}.$$

*Then*

$$\begin{aligned} A(t,T) &= S_t - I_t, \\ f_t(K,T) &= S_t - I_t - Ke^{-r(T-t)}, \\ F(t,T) &= (S_t - I_t)e^{r(T-t)} \\ &= S_t e^{r(T-t)} - \sum_{i=1}^{m} d_i e^{r(T-t_i)}. \end{aligned}$$

*Proof.* Buy one share and borrow $d_i e^{-r(t_i-t)}$ separately for each dividend date $t_i$. At $t_i$, the corresponding debt is

$$d_i e^{-r(t_i-t)} e^{r(t_i-t)} = d_i,$$

which is repaid using that dividend. There are no net intermediate cash flows and one share remains at $T$. The initial cost is $S_t - I_t$, proving the prepaid-forward formula. Borrowing an additional $Ke^{-r(T-t)}$ changes the terminal payoff to $S_T - K$, yielding the forward-value formula. Finally,

$$\begin{aligned} F(t,T) &= \left( S_t - \sum_{i=1}^{m} d_i e^{-r(t_i-t)} \right) e^{r(T-t)} \\ &= S_t e^{r(T-t)} - \sum_{i=1}^{m} d_i e^{r[(T-t)-(t_i-t)]} \\ &= S_t e^{r(T-t)} - \sum_{i=1}^{m} d_i e^{r(T-t_i)}. \end{aligned}$$

□

**Alternative cash-and-carry derivation**

Borrow $S_t$, buy one share, reinvest every cash dividend in the bank, and short a zero-cost forward with delivery price $K$. Terminal wealth is

$$\begin{aligned} Z_T &= S_T + \sum_i d_i e^{r(T-t_i)} - S_t e^{r(T-t)} + (K - S_T) \\ &= K - S_t e^{r(T-t)} + \sum_i d_i e^{r(T-t_i)}. \end{aligned}$$

This quantity is deterministic. If it is positive, use the strategy; if negative, reverse it. No-arbitrage therefore forces it to equal zero.

**Example 1.3.** Let $S_0 = 100$, $r = 0.05$, $T = 1$, and let the only dividend be two dollars at $t_1 = 0.5$. Then

$$\begin{aligned} F(0,1) &= 100e^{0.05} - 2e^{0.05(1-0.5)} \\ &\approx 103.0765. \end{aligned}$$

Subtracting merely two dollars from the no-dividend forward price would omit the interest earned on the dividend between 0.5 and 1.

*Remark* 1.11 (Uncertain dividends). If the dividend amounts are random and not otherwise hedged, the displayed terminal sum is random. No-arbitrage does not let us set it equal to a fixed delivery price state by state. Replacing those dividends by physical expectations is not a model-free pricing argument. Additional tradable claims or modelling assumptions are needed.

### 1.3.5 Futures, marking to market, and margin

**Definition 1.14** (Futures settlement)**.** A futures contract is standardized and exchange-traded, with gains and losses settled through marking to market. If futures quotes at settlement dates $t_0 < \cdots < t_n = T$ are $G_0, \ldots, G_n$, a unit long position receives

$$G_i - G_{i-1}$$

at date $t_i$. For a cash-settled contract on the terminal spot value, $G_n = S_T$.

The undiscounted sum telescopes:

$$\begin{aligned} \sum_{i=1}^{n} (G_i - G_{i-1}) &= G_n - G_0 \\ &= S_T - G_0. \end{aligned}$$

But if each settlement is financed or invested until $T$, total terminal wealth is instead

$$\sum_{i=1}^{n} (G_i - G_{i-1}) e^{r(T-t_i)}.$$

These are different expressions. A forward makes one payment $S_T - K$ at $T$; a futures position produces a stream of payments before $T$.

**Example 1.4** (Settlement timing matters)**.** Suppose $G_0 = 100$, $G_1 = 110$ at $t_1 = 0.5$, $G_2 = 105$ at $T = 1$, and $r = 0.05$. Then the long futures position receives 10 and later pays 5. Its accumulated settlement value is

$$10e^{0.05(0.5)} - 5 \approx 5.2532,$$

whereas a forward entered at $K = 100$ pays $105 - 100 = 5$. This comparison isolates settlement timing; it does not assert that these particular starting quotes are fair in a specified model.

**Proposition 1.4** (Why deterministic-rate futures and forward quotes agree)**.** *In an idealized settlement model with deterministic financing rates, unrestricted fractional futures holdings, and $G_n = S_T$, the initial futures quote equals the forward delivery price for the same terminal underlying and delivery date.*

*Proof.* During $(t_{i-1}, t_i]$, hold $q_i = e^{-r(T-t_i)}$ futures contracts. Changes to the number of contracts at settlement have zero contract value; all gains and losses are put in the bank account, allowing borrowing when needed. The terminal value of settlement $i$ is

$$q_i(G_i - G_{i-1})e^{r(T-t_i)} = G_i - G_{i-1}.$$

Summing produces $S_T - G_0$ from zero initial wealth. This replicates a zero-value forward with delivery price $G_0$. Hence $G_0 = F(t_0, T)$. □

*Remark* 1.12. With random rates, the required future accumulation factor need not be known at $t_{i-1}$. The argument fails, and equality is not automatic. Margin funding frictions can also affect the practical comparison.

**Definition 1.15** (Margin)**.** Initial margin is collateral required to support a futures position. Variation margin transfers settlement gains and losses. A margin call requests additional collateral or funding when the account falls below the required level.

Zero contract value at inception does not mean zero cash resources are required. Margin is distinct from an option premium or an asset purchase price. A hedge can lose cash before it delivers its intended economic offset, creating a funding requirement.

*Remark* 1.13 (Institutional descriptions). Forwards are commonly privately negotiated, while futures use standardized exchange and clearing arrangements. Clearing and collateral mitigate default exposure rather than making it logically impossible. Being over the counter does not mean being unregulated. These institutional remarks motivate the model; they are not used to prove the pricing identities.

## 1.4 Options: rights, payoffs, and profit

*Source coverage: lecture slides 41–53, 55–57..*

### 1.4.1 Calls, puts, and exercise styles

**Definition 1.16** (Call and put)**.** A call gives its holder the right, but not the obligation, to buy one unit of the underlying for $K$. A put gives its holder the right, but not the obligation, to sell one unit for $K$. The number $K$ is the strike price, and $T$ is the expiry date.

**Definition 1.17** (Exercise styles)**.** A European option can be exercised only at $T$. An American option can be exercised at any permitted time in $[t, T]$. A Bermudan option can be exercised on a specified set of dates, including its final expiry date.

The long holder chooses whether to exercise. The short writer must honour that choice. These names describe exercise rights, not geographical trading locations.

**Definition 1.18** (Exercise time)**.** In a stochastic formulation, an American exercise rule is a stopping time $\tau$ with values in $[t, T]$: for every $u$, the event $\{\tau \le u\}$ belongs to $\mathcal{F}_u$. Thus the decision cannot use information that arrives only later.

No optimal-stopping pricing theorem is needed here. To prove an American upper bound, we construct a hedge that works for every permitted $\tau$.

### 1.4.2 The positive part and elementary payoff identities

**Lemma 1.2** (Positive-part identities)**.** *For every $x \in \mathbb{R}$,*

$$x^+ = \begin{cases} x, & x \ge 0, \\ 0, & x < 0, \end{cases}$$
$$x^+ - (-x)^+ = x,$$
$$x^+ + (-x)^+ = |x|.$$

*Moreover, $x \mapsto x^+$ is increasing, 1-Lipschitz, and convex.*

*Proof.* If $x \ge 0$, the two positive parts are $x$ and zero; if $x < 0$, they are zero and $-x$. This proves the identities. For $x \le y$, check three cases:

$$\begin{aligned} y \le 0: &\quad y^+ - x^+ = 0, \\ x \le 0 \le y: &\quad y^+ - x^+ = y \le y - x, \\ 0 \le x \le y: &\quad y^+ - x^+ = y - x. \end{aligned}$$

Thus $0 \le y^+ - x^+ \le y - x$, proving monotonicity and the Lipschitz bound. For $\lambda \in [0, 1]$, both $x \le x^+$ and $y \le y^+$ imply

$$\lambda x + (1 - \lambda)y \le \lambda x^+ + (1 - \lambda)y^+.$$

The right side is nonnegative, so taking the maximum of zero and the left side preserves the inequality:

$$(\lambda x + (1 - \lambda)y)^+ \le \lambda x^+ + (1 - \lambda)y^+.$$

This is convexity. □

**Proposition 1.5** (Vanilla terminal payoffs)**.** *For a stock price $s \geq 0$ at exercise, the long call and long put payoffs are*

$$h_c(s) = (s-K)^+ = \begin{cases} 0, & s \leq K, \\ s-K, & s > K, \end{cases}$$

$$h_p(s) = (K-s)^+ = \begin{cases} K-s, & s < K, \\ 0, & s \geq K. \end{cases}$$

*Short positions have payoffs $-h_c(s)$ and $-h_p(s)$.*

*Proof.* A call holder can buy for $K$ and acquire value $s$, producing net value $s-K$, or decline to exercise and receive zero. Choose the larger. A put holder can deliver value $s$ for $K$, producing $K-s$, or receive zero. The writer pays the holder's contractual receipt. □

*Remark* 1.14 (A short call is not a long put)*.* A short call pays $-(s-K)^+$, whereas a long put pays $(K-s)^+$. Their values coincide only in some states. In particular, a short call has unbounded loss as $s \to \infty$, while a long put has nonnegative payoff bounded by $K$.

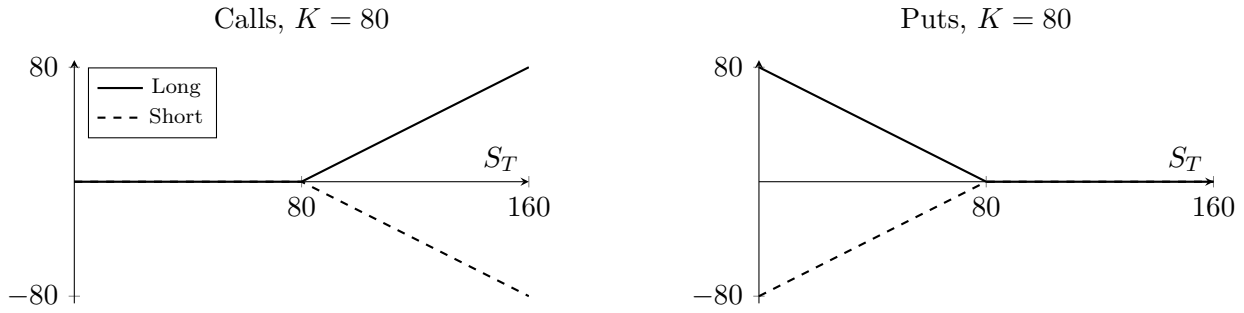



Figure 1.1: Terminal payoffs before accounting for the premium. Solid lines are long positions; dashed lines are short positions.

### 1.4.3 Moneyness, intrinsic value, and exercise

**Definition 1.19** (Moneyness)**.** A call is in the money if $S_t > K$, at the money if $S_t = K$, and out of the money if $S_t < K$. A put is in the money if $S_t < K$, at the money if $S_t = K$, and out of the money if $S_t > K$.

The intrinsic values are $(S_t - K)^+$ and $(K - S_t)^+$. An out-of-the-money option has zero intrinsic value, not a negative optimal exercise payoff: the holder can decline an unfavourable exchange. European moneyness describes the current spot-strike comparison even though immediate exercise is unavailable.

**Definition 1.20** (Option price notation)**.** Write $c_t = c(K,t,T)$ and $p_t = p(K,t,T)$ for European call and put prices, and $C_t = C(K,t,T)$ and $P_t = P(K,t,T)$ for American call and put prices.

**Proposition 1.6** (More exercise rights cannot reduce value)**.** *For otherwise identical contracts,*

$$C_t \geq c_t, \qquad P_t \geq p_t.$$

*If a Bermudan option includes $T$ among its exercise dates, its value lies between the corresponding European and American values. Also,*

$$C_t \geq (S_t - K)^+, \qquad P_t \geq (K - S_t)^+.$$

*Proof.* If the contract with more exercise rights were cheaper, buy it and sell the restricted contract. Exercise the long contract whenever the short contract is exercised, so the obligations cancel. Invest the initial receipt. This is arbitrage. For the intrinsic-value bounds, a holder can exercise immediately, or choose not to exercise and obtain at least zero. A price below either achievable nonnegative immediate receipt would give an immediate arbitrage. □

### 1.4.4 Payoff versus financed profit

**Definition 1.21** (Terminal financed profit)**.** For a portfolio costing $V_t$ at $t$, with terminal payoff $H_T$ and no intermediate payments, its terminal profit relative to risk-free financing is

$$\Pi_T = H_T - V_t e^{r(T-t)}.$$

The corresponding discounted profit is $D(t,T)H_T - V_t$.

For long and short calls,

$$\Pi_T^{\text{long call}} = (S_T - K)^+ - c_t e^{r(T-t)},$$
$$\Pi_T^{\text{short call}} = c_t e^{r(T-t)} - (S_T - K)^+.$$

The long call can lose its entire premium plus its financing cost. The short call receives a premium initially but retains unbounded potential loss. For a put, replace $(S_T - K)^+$ by $(K - S_T)^+$.

**Example 1.5** (The lecture's call-profit calculation)**.** Take $K = 260$ and premium $c_0 = 1.08$ per share. Ignoring financing, profit is

$$\begin{aligned}\Pi_T &= (S_T - 260)^+ - 1.08 \\ &= \begin{cases} -1.08, & S_T \leq 260, \\ S_T - 261.08, & S_T > 260. \end{cases}\end{aligned}$$

Thus the break-even price is 261.08. The simple premium return is

$$R = \frac{(S_T - 260)^+ - 1.08}{1.08}.$$

For $S_T \leq 260$, this is $-1$, or a loss of 100%. Including financing changes break-even to $260 + 1.08e^{rT}$. For a contract covering $m$ shares, multiply all dollar amounts by $m$; break-even per share is unchanged.

*Remark* 1.15 (Reading an option chain). An option chain organizes quotes by strike and expiry, usually separating calls and puts. A bid is a price at which one can sell, while an ask is a price at which one can buy. Mixing asks, stale trades, or nonsynchronous observations does not produce a single frictionless price system. The mathematical inequalities below apply to consistently executable prices under the standing assumptions.

## 1.5 European and American option price bounds

*Source coverage: lecture slides 54–68..*

### 1.5.1 European bounds from prepaid delivery

Set

$$A = A(t,T) = S_t e^{-\delta(T-t)}, \qquad D = D(t,T) = e^{-r(T-t)}.$$

The asset costing $A$ delivers one share at $T$ and the asset costing $KD$ delivers $K$ dollars at $T$.

**Theorem 1.5** (European vanilla bounds)**.** *For a nonnegative underlying with constant dividend yield,*

$$\begin{aligned} \max\{A - KD, 0\} \le c_t \le A, \\ \max\{KD - A, 0\} \le p_t \le KD. \end{aligned}$$

*These inequalities do not require $r \ge 0$.*

*Proof.* For every $s \ge 0$ and $K \ge 0$,

$$\begin{aligned} 0 \le (s-K)^+ \le s, \\ s - K \le (s-K)^+, \\ 0 \le (K-s)^+ \le K, \\ K - s \le (K-s)^+. \end{aligned}$$

The claims $S_T$, $K$, $S_T - K$, and $K - S_T$ have prices $A$, $KD$, $A - KD$, and $KD - A$. Apply payoff dominance to each comparison:

$$\begin{array}{lll} c_t \ge 0, & c_t \ge A - KD, & c_t \le A, \\ p_t \ge 0, & p_t \ge KD - A, & p_t \le KD. \end{array}$$

Combining the lower bounds gives the result. □

**The call lower bound as a cash-flow comparison**

Compare a call with a long prepaid share and a loan of $KD$:

| Portfolio | Cost at $t$ | $S_T < K$ | $S_T \ge K$ |
|---|---|---|---|
| Call | $c_t$ | $0$ | $S_T - K$ |
| Prepaid share minus bond | $A - KD$ | $S_T - K$ | $S_T - K$ |

The call dominates in every state, so it cannot be cheaper. For a put, reverse the prepaid share and bond: its payoff dominates $K - S_T$.

*Remark* 1.16 (Sharper upper bounds). The lecture uses $c_t \le S_t$ and, for $r \ge 0$, $p_t \le K$. The prepaid-share and discounted-bond arguments give the sharper European bounds $c_t \le A$ and $p_t \le KD$. When $\delta \ge 0$ and $r \ge 0$, these imply the lecture's bounds.

**Example 1.6** (Checking a quote)**.** Let $S_t = 100$, $K = 105$, $T - t = 0.5$, $r = 0.04$, and $\delta = 0.02$. Then

$$\begin{aligned} A &= 100e^{-0.01} \approx 99.0050, \\ KD &= 105e^{-0.02} \approx 102.9209, \\ A - KD &\approx -3.9159. \end{aligned}$$

Thus $0 \le c_t \le 99.0050$ and $3.9159 \le p_t \le 102.9209$. These bounds are necessary; they do not identify a unique option value.

### 1.5.2 American upper bounds

**Theorem 1.6** (American upper and lower bounds)**.** *Assume $r \ge 0$ and $\delta \ge 0$. Then*

$$\begin{aligned} \max\{c_t, (S_t - K)^+\} \le C_t \le S_t, \\ \max\{p_t, (K - S_t)^+\} \le P_t \le K. \end{aligned}$$

*In particular,*

$$\begin{aligned} S_t \ge C_t \ge c_t \ge \max\{A - KD, 0\}, \\ K \ge P_t \ge p_t \ge \max\{KD - A, 0\}. \end{aligned}$$

*Proof.* The lower bounds were proved from exercise-right inclusion and immediate exercise.

For the call upper bound, buy one share and sell one American call. If the call is exercised at $\tau$, deliver the share and receive $K \ge 0$. Dividends received before $\tau$ are nonnegative and can be banked. If the option expires without exercise, the remaining share and banked dividends have nonnegative value. Therefore this covered position has nonnegative remaining wealth under every exercise decision. If $C_t > S_t$, its setup produces a strictly positive receipt as well, giving arbitrage. Hence $C_t \le S_t$.

For the put upper bound, deposit $K$ in the bank account and sell one American put. At any exercise time $\tau$, the deposit has value $Ke^{r(\tau-t)} \ge K$. Pay $K$ and receive a share worth $S_\tau \ge 0$. Any remaining cash and stock have nonnegative value. If no exercise occurs, keep the bank account. If $P_t > K$, this superhedge also produces an initial receipt, giving arbitrage. Thus $P_t \le K$. □

*Remark* 1.17 (Where the rate assumption enters). The put proof needs $Ke^{r(\tau-t)} \ge K$. For negative rates, holding $K$ initially may fail to cover a later $K$ payment. A safe general bound for constant $r$ is

$$P_t \le K \max\{1, e^{-r(T-t)}\}.$$

Indeed this initial deposit is at least $Ke^{-r(\tau-t)}$ for every $\tau \in [t, T]$. Similarly, the call hedge assumes the stock position does not require funding negative distributions; nonnegative dividends are sufficient.

### 1.5.3 Why a non-dividend-paying American call has no early-exercise premium

**Theorem 1.7** (American call equals European call)**.** *If the stock pays no dividends and $r \geq 0$, then*

$$C(K,t,T) = c(K,t,T).$$

*Early exercise never gives a strictly better value than retaining the option's remaining exercise rights. If $r > 0$, $K > 0$, and $u < T$, exercise of an in-the-money call is strictly dominated by selling the option at its arbitrage-free price.*

*Proof.* First, at any time $u < T$, the European lower bound implies

$$\begin{aligned} c(K,u,T) &\geq S_u - Ke^{-r(T-u)} \\ &= S_u - K + K\left(1 - e^{-r(T-u)}\right) \\ &\geq S_u - K. \end{aligned}$$

It also implies $c(K,u,T) \geq 0$. Consequently,

$$c(K,u,T) \geq (S_u - K)^+.$$

We already know $C_t \geq c_t$. To establish the reverse inequality rigorously, suppose $C_t > c_t$. Buy the European call, sell the American call, and invest the initial receipt $C_t - c_t$. If the short American call is exercised at $\tau < T$, sell the European call. Its market value satisfies

$$c(K,\tau,T) \geq (S_\tau - K)^+,$$

so the sale funds the short contract's exercise obligation. For physical settlement, the stock can be bought at $S_\tau$ and delivered against receipt of $K$; the net cost is $S_\tau - K$, also covered by the same bound. Bank any surplus. If exercise is deferred to $T$, the two terminal payoffs cancel. In either case the invested initial receipt remains strictly positive. This would be arbitrage, so $C_t \leq c_t$.

For $r > 0$, $K > 0$, and $u < T$, the term $K(1 - e^{-r(T-u)})$ is strictly positive. If $S_u > K$, the option's price is therefore strictly above the exercise receipt $S_u - K$. □

*Remark* 1.18 (Weak dominance and investor preferences)*.* At $r = 0$ or $K = 0$, indifference can occur, so "never exercise early" should be read as "early exercise is not strictly necessary to obtain the option's value". An investor may still wish to sell the option before expiry. This theorem concerns the value of the exercise right, not an instruction to maintain market exposure until $T$.

*Remark* 1.19 (Why dividends change the conclusion)*.* For $\delta > 0$, the lower bound becomes $S_u e^{-\delta(T-u)} - Ke^{-r(T-u)}$, which need not exceed $S_u - K$. Exercising acquires the stock and its future dividend rights. Therefore early exercise can have value; positive dividends do not imply that early exercise is always optimal.

**Example 1.7** (Negative rates invalidate the no-early-exercise conclusion)**.** Consider a deterministic non-dividend stock $S_u = S_t e^{r(u-t)}$ with $r < 0$, and suppose $S_T > K$. The European payoff is deterministic, so

$$\begin{aligned} c_t &= e^{-r(T-t)}(S_t e^{r(T-t)} - K) \\ &= S_t - Ke^{-r(T-t)} \\ &< S_t - K. \end{aligned}$$

Immediate American exercise is worth $S_t - K$, exceeding the European value. This market has no stock-versus-bank arbitrage: the stock grows at the same rate as the bank account.

### 1.5.4 American puts and early exercise

**Example 1.8** (Early exercise can be optimal for a put)**.** Suppose the stock is worthless now and remains zero, and $r > 0$, $K > 0$. Exercising an American put now gives $K$. Waiting until $T$ gives a payment with present value $Ke^{-r(T-t)} < K$. Thus

$$\begin{aligned} P_t &= K, \\ p_t &= Ke^{-r(T-t)}, \\ P_t - p_t &= K(1 - e^{-r(T-t)}) > 0. \end{aligned}$$

This proves that an American put can have a strictly positive early-exercise premium. It does not prove that every American put should be exercised early.

**Proposition 1.7** (No put exercise premium at zero rates under nonnegative yield)**.** *If $r = 0$ and $\delta \geq 0$, then $P_t = p_t$.*

*Proof.* At every possible exercise time $u$, the European put lower bound gives

$$\begin{aligned} p(K, u, T) &\geq \max\{K - S_u e^{-\delta(T-u)}, 0\} \\ &\geq (K - S_u)^+. \end{aligned}$$

A long European put can therefore be sold to cover any exercise obligation on a short American put. If $P_t > p_t$, buy the European, sell the American, and invest the difference to obtain arbitrage, exactly as in the call proof. Since $P_t \geq p_t$, equality follows. □

*Remark* 1.20. The lecture's statement about possible early put exercise should not be interpreted as a universal optimal-exercise rule. Positive rates encourage earlier receipt of the strike, while retaining optionality and receiving the benefit of future price declines can encourage waiting. Part I supplies comparisons, not a general exercise boundary.

## 1.6 Put-call parity and its American bounds

*Source coverage: lecture slides 69–76..*

### 1.6.1 European parity

**Theorem 1.8** (Put-call parity)**.** *For European options on the same underlying with the same strike $K$ and maturity $T$,*

$$\boxed{c_t - p_t = A(t,T) - KD(t,T).}$$

*With continuous proportional dividends,*

$$c_t - p_t = S_t e^{-\delta(T-t)} - Ke^{-r(T-t)}.$$

*Proof.* At maturity,

$$\begin{aligned}(S_T - K)^+ - (K - S_T)^+ &= \begin{cases}(S_T - K) - 0, & S_T \geq K,\\ 0 - (K - S_T), & S_T < K\end{cases}\\ &= S_T - K.\end{aligned}$$

Thus long call minus long put replicates long prepaid share minus a bond paying $K$. The two costs must coincide:

$$c_t - p_t = A(t,T) - KD(t,T).$$

□

Equivalently,

$$\begin{aligned}c_t + KD &= p_t + A,\\ c_t &= p_t + A - KD,\\ p_t &= c_t - A + KD.\end{aligned}$$

The first identity says that a call plus enough cash to pay the strike is equivalent to a prepaid share protected by a put.

**Corollary 1.2** (Synthetic forward)**.** *A long call and short put with common $K, T$ have exactly the value and payoff of the long forward with delivery price $K$:*

$$c_t - p_t = f_t(K,T).$$

*This synthetic forward has zero cost only if $K = F(t,T)$.*

**Corollary 1.3** (Discrete-dividend parity)**.** *For known discrete dividends with present value $I_t$,*

$$c_t - p_t = S_t - I_t - KD(t,T).$$

*Proof.* The payoff identity is unchanged; replace the prepaid-share price by $S_t - I_t$. □

### 1.6.2 Explicit arbitrage for a parity violation

Let

$$\varepsilon = (c_t - p_t) - (A - KD).$$

If $\varepsilon > 0$, sell the call, buy the put, buy prepaid stock, and borrow $KD$. The setup cash flow is

$$c_t - p_t - A + KD = \varepsilon > 0,$$

and the terminal cash flow before banking this receipt is

$$\begin{aligned} -(S_T - K)^+ + (K - S_T)^+ + S_T - K &= -(S_T - K) + S_T - K \\ &= 0. \end{aligned}$$

Investing the initial receipt yields terminal profit $\varepsilon e^{r(T-t)}$. If $\varepsilon < 0$, reverse every position and bank $-\varepsilon$.

**Example 1.9** (The lecture's numerical parity trade)**.** Treat the quoted instruments as European and assume no dividends. Use the lecture's stylized data

$$S_0 = 245, \quad K = 250, \quad T = \frac{1}{52}, \quad r = 0.02, \quad c_0 = 3.20, \quad p_0 = 8.10.$$

Then

$$\begin{aligned} c_0 - p_0 &= -4.90, \\ S_0 - Ke^{-rT} &= 245 - 250e^{-0.02/52} \\ &\approx -4.903864643, \\ \varepsilon &= -4.90 - (-4.903864643) \\ &\approx 0.003864643 > 0. \end{aligned}$$

Sell one call, buy one put, buy one share, and borrow $250e^{-0.02/52}$. The complete cash-flow table is

| Position | At 0 | At $T$, $S_T < 250$ | At $T$, $S_T \geq 250$ |
|---|---|---|---|
| Buy stock | $-245$ | $S_T$ | $S_T$ |
| Borrow | $+249.903864643$ | $-250$ | $-250$ |
| Sell call | $+3.20$ | $0$ | $-(S_T - 250)$ |
| Buy put | $-8.10$ | $250 - S_T$ | $0$ |
| Total | $+0.003864643$ | $0$ | $0$ |

Banking the receipt makes the strategy zero-cost and produces approximately 0.003866129 at $T$ per share of exposure.

*Remark* 1.21 (The example's modelling status). The slide data are historical and rounded. The exact-parity trade assumes European exercise, synchronous executable prices, no dividends, and accessible 2% financing. American equity-option quotes cannot simply be substituted into European parity without checking exercise features. A tiny algebraic discrepancy in indicative quotes is not evidence of an executable real-market arbitrage.

### 1.6.3 American put-call inequalities

**Theorem 1.9** (American parity bounds without dividends)**.** *Assume no dividends and $r \geq 0$. Then*

$$\boxed{S_t - K \leq C_t - P_t \leq S_t - KD(t,T).}$$

*Proof.* For the upper bound, $C_t = c_t$ and $P_t \geq p_t$. Therefore

$$\begin{aligned} C_t - P_t &\leq c_t - p_t \\ &= S_t - KD(t,T). \end{aligned}$$

For the lower bound, construct a portfolio consisting of one long American call, $K$ dollars in the bank, one short American put, and one short share. Its setup cost is

$$C_t + K - P_t - S_t.$$

If the short put is exercised at time $\tau$, pay $K$ from the bank and receive the share. Use that share to close the short stock position. The cash remaining is

$$Ke^{r(\tau - t)} - K \geq 0,$$

and the unexercised long call has nonnegative value. If the put is not exercised early, at maturity the call, put, and stock positions combine as

$$\begin{aligned} (S_T - K)^+ - (K - S_T)^+ - S_T &= (S_T - K) - S_T \\ &= -K. \end{aligned}$$

The bank account covers this with $Ke^{r(T-t)} \geq K$. Thus the portfolio never requires outside funding and has nonnegative residual value. A negative initial cost would yield arbitrage after banking the receipt. Hence

$$\begin{aligned} C_t + K - P_t - S_t &\geq 0, \\ C_t - P_t &\geq S_t - K. \end{aligned}$$

□

**Corollary 1.4** (Bound on the put's early-exercise premium)**.** *Under the same assumptions,*

$$0 \leq P_t - p_t \leq K(1 - D(t,T)).$$

*Proof.* Use $C_t = c_t$, $c_t - p_t = S_t - KD$, and $C_t - P_t \geq S_t - K$:

$$\begin{aligned} S_t - KD - (P_t - p_t) &\geq S_t - K, \\ P_t - p_t &\leq K - KD. \end{aligned}$$

The lower bound follows from the American exercise advantage. □

*Remark* 1.22 (The gap is not universally small). The width of the American parity interval is $K(1 - e^{-r(T-t)})$. It is approximately $Kr(T-t)$ for a small product $r(T-t)$, but it can be economically large for long maturities, high rates, or large strikes.

## 1.7 Dependence on strike: monotonicity, Lipschitz bounds, and convexity

*Source coverage: lecture slides 77–86..* Throughout this section, fix $t, T$, the underlying, and the exercise style. Abbreviate $c(K) = c(K, t, T)$, and similarly for $p, C, P$. All comparisons require the same underlying and maturity.

### 1.7.1 European strike spreads

**Lemma 1.3** (Pointwise strike-difference bounds)**.** *For $0 \le K_1 \le K_2$, let $d = K_2 - K_1$. For every $s \ge 0$,*

$$0 \le (s - K_1)^+ - (s - K_2)^+ \le d,$$
$$0 \le (K_2 - s)^+ - (K_1 - s)^+ \le d.$$

*Proof.* The call difference is

$$(s - K_1)^+ - (s - K_2)^+ = \begin{cases} 0, & s \le K_1, \\ s - K_1, & K_1 < s < K_2, \\ K_2 - K_1, & s \ge K_2. \end{cases}$$

Every branch lies between zero and $d$. The put difference is

$$(K_2 - s)^+ - (K_1 - s)^+ = \begin{cases} K_2 - K_1, & s \le K_1, \\ K_2 - s, & K_1 < s < K_2, \\ 0, & s \ge K_2, \end{cases}$$

and the same bounds follow. □

**Theorem 1.10** (European monotonicity and discounted Lipschitz bounds)**.** *For $K_1 \le K_2$,*

$$0 \le c(K_1) - c(K_2) \le D(t, T)(K_2 - K_1),$$
$$0 \le p(K_2) - p(K_1) \le D(t, T)(K_2 - K_1).$$

*Thus European calls are nonincreasing in strike, and European puts are nondecreasing in strike. Both are $D(t, T)$-Lipschitz.*

*Proof.* Each difference portfolio has terminal payoff between zero and $d$. The price comparison principle places its price between zero and $dD(t, T)$. □

*Remark* 1.23 (Recovering the lecture's 1-Lipschitz statement). If $r \ge 0$, then $D(t, T) \le 1$. Consequently,

$$|c(K_2) - c(K_1)| \le |K_2 - K_1|,$$
$$|p(K_2) - p(K_1)| \le |K_2 - K_1|.$$

For negative $r$, the discounted constant can exceed one, so the 1-Lipschitz claim is not generally valid. The discounted bounds remain valid.

### 1.7.2 American strike comparisons: every exercise time must be covered

**Theorem 1.11** (American monotonicity and 1-Lipschitz bounds)**.** *Assume $r \geq 0$. For $K_1 \leq K_2$,*

$$0 \leq C(K_1) - C(K_2) \leq K_2 - K_1,$$
$$0 \leq P(K_2) - P(K_1) \leq K_2 - K_1.$$

*Proof.* Write $d = K_2 - K_1$. We prove each inequality with a covering portfolio.

**Call monotonicity.** Buy a call at $K_1$ and sell a call at $K_2$. If the short option is exercised at $\tau$, the pointwise inequality

$$(S_\tau - K_1)^+ \geq (S_\tau - K_2)^+$$

shows that the long option can cover its obligation by exercise. If the short expires unexercised, the long position still has nonnegative payoff. Therefore $C(K_1) - C(K_2) \geq 0$.

**Call spread upper bound.** Buy the call at $K_2$, sell the call at $K_1$, and place $d$ dollars in the bank. At the exercise time $\tau$ of the short option, the bank has $de^{r(\tau-t)} \geq d$. The pointwise spread bound implies

$$\begin{aligned}(S_\tau - K_2)^+ + de^{r(\tau-t)} &\geq (S_\tau - K_2)^+ + d \\ &\geq (S_\tau - K_1)^+.\end{aligned}$$

Exercise the long option if useful and use the cash to settle the short. If the short is never exercised, the remaining assets are nonnegative. Therefore the superhedge cannot have negative cost:

$$C(K_2) + d - C(K_1) \geq 0,$$
$$C(K_1) - C(K_2) \leq d.$$

**Put monotonicity.** Buy a put at $K_2$ and sell a put at $K_1$. At any exercise time of the short,

$$(K_2 - S_\tau)^+ \geq (K_1 - S_\tau)^+.$$

Thus the long covers the short and $P(K_2) - P(K_1) \geq 0$.

**Put spread upper bound.** Buy a put at $K_1$, sell a put at $K_2$, and bank $d$. At exercise,

$$\begin{aligned}(K_1 - S_\tau)^+ + de^{r(\tau-t)} &\geq (K_1 - S_\tau)^+ + d \\ &\geq (K_2 - S_\tau)^+.\end{aligned}$$

The same covering argument gives $P(K_1) + d - P(K_2) \geq 0$, or $P(K_2) - P(K_1) \leq d$. □

*Remark* 1.24 (Cash settlement is a convenient representation)*.* A physical call exercise can be settled by buying a share at $S_\tau$, delivering it, and receiving $K$; the net cost is $S_\tau - K$. A physical put exercise can be settled by paying $K$, receiving the share, and selling it for $S_\tau$; the net cost is $K - S_\tau$. These transactions justify using exercise cash values in the covering arguments. An irrational out-of-the-money exercise by the short holder benefits the writer and cannot invalidate the bounds.

*Remark* 1.25 (A negative-rate extension)*.* For constant $r < 0$, deposit $de^{-r(T-t)}$ rather than $d$. This covers a payment of at most $d$ at any exercise date. Thus an always-valid bound in this financing model is

$$|C(K_2) - C(K_1)|,\ |P(K_2) - P(K_1)| \leq d \max\{1, D(t,T)\}.$$

Monotonicity itself does not require nonnegative rates.

### 1.7.3 Convexity in the strike

**Definition 1.22** (Convex function)**.** A function $g$ on an interval is convex if, for every $x, y$ in that interval and $\lambda \in [0,1]$,

$$g(\lambda x + (1-\lambda)y) \le \lambda g(x) + (1-\lambda)g(y).$$

Geometrically, the graph lies below each chord joining two of its points.

**Theorem 1.12** (Convexity of vanilla prices in strike)**.** *Each of $c, p, C, P$ is convex as a function of strike. In particular, writing $\bar{K} = \lambda K_1 + (1-\lambda)K_2$,*

$$\begin{aligned} c(\bar{K}) &\le \lambda c(K_1) + (1-\lambda)c(K_2), \\ p(\bar{K}) &\le \lambda p(K_1) + (1-\lambda)p(K_2), \\ C(\bar{K}) &\le \lambda C(K_1) + (1-\lambda)C(K_2), \\ P(\bar{K}) &\le \lambda P(K_1) + (1-\lambda)P(K_2). \end{aligned}$$

*Proof.* For a fixed stock value $s$,

$$\begin{aligned} (s-\bar{K})^+ &= \big(\lambda(s-K_1) + (1-\lambda)(s-K_2)\big)^+ \\ &\le \lambda(s-K_1)^+ + (1-\lambda)(s-K_2)^+, \\ (\bar{K}-s)^+ &= \big(\lambda(K_1-s) + (1-\lambda)(K_2-s)\big)^+ \\ &\le \lambda(K_1-s)^+ + (1-\lambda)(K_2-s)^+. \end{aligned}$$

For European options, buy the indicated weighted endpoint options and sell one option at $\bar{K}$. The terminal payoff is nonnegative, so the initial cost is nonnegative. This proves the first two inequalities.

For American options, buy the same weighted endpoint options and sell the middle-strike option. At the exercise time $\tau$ of the short option, exercise those long positions with positive exercise value. The pointwise inequalities, with $s = S_\tau$, show that their combined receipts cover the obligation. Any unused long rights can be retained or discarded at nonnegative value. If the short is not exercised before expiry, the same comparison works at $T$. Thus the portfolio is a superhedge under every exercise time, so its cost is nonnegative. This proves the remaining two inequalities. □

**Explicit piecewise call-butterfly calculation**

Assume $K_1 < K_2$ and $0 < \lambda < 1$, so $K_1 < \bar{K} < K_2$. The weighted payoff

$$b(s) = \lambda(s-K_1)^+ + (1-\lambda)(s-K_2)^+ - (s-\bar{K})^+$$

is

$$b(s) = \begin{cases} 0, & s \le K_1, \\ \lambda(s-K_1), & K_1 < s < \bar{K}, \\ (1-\lambda)(K_2-s), & \bar{K} \le s < K_2, \\ 0, & s \ge K_2. \end{cases}$$

For the third branch, the algebra is

$$\begin{aligned} \lambda(s-K_1) - (s-\bar{K}) &= \lambda s - \lambda K_1 - s + \lambda K_1 + (1-\lambda)K_2 \\ &= (1-\lambda)(K_2-s) \ge 0. \end{aligned}$$

For the fourth branch,

$$\begin{aligned}\lambda(s-K_1)+(1-\lambda)(s-K_2)-(s-\bar{K}) &= s-\lambda K_1-(1-\lambda)K_2-s+\bar{K}\\ &= 0.\end{aligned}$$

The put version has exactly the same weighted payoff: use $(K-s)^+=(s-K)^++K-s$, and observe that the affine strike terms cancel.

### 1.7.4 The lecture's convexity arbitrage

**Example 1.10** (Unequally spaced strikes)**.** Suppose European call quotes with common maturity are

$$c(50)=14, \qquad c(59)=8.9, \qquad c(65)=5.$$

Solve for the endpoint weight:

$$\begin{aligned}59 &= \lambda 50+(1-\lambda)65,\\ 15\lambda &= 6,\\ \lambda &= 0.4.\end{aligned}$$

Convexity requires

$$\begin{aligned}c(59) &\le 0.4c(50)+0.6c(65)\\ &= 0.4(14)+0.6(5)=8.6.\end{aligned}$$

The quoted 8.9 violates this bound. Buy four calls at 50, buy six calls at 65, and sell ten calls at 59. The initial cash flow is

$$\begin{aligned}-4(14)-6(5)+10(8.9) &= -56-30+89\\ &= 3.\end{aligned}$$

The terminal payoff is

$$\begin{aligned}H(s) &= 4(s-50)^+ + 6(s-65)^+ - 10(s-59)^+\\ &= \begin{cases}0, & s\le 50,\\ 4(s-50), & 50<s\le 59,\\ 4(s-50)-10(s-59), & 59<s<65,\\ 4(s-50)+6(s-65)-10(s-59), & s\ge 65\end{cases}\\ &= \begin{cases}0, & s\le 50,\\ 4(s-50), & 50<s\le 59,\\ 6(65-s), & 59<s<65,\\ 0, & s\ge 65.\end{cases}\end{aligned}$$

This is nonnegative everywhere and reaches 36 at $s=59$. Bank the initial three-dollar receipt. The zero-cost strategy then has wealth $H(S_T)+3e^{r(T-t)}>0$ in every state.

### 1.7.5 Slopes, continuity, and discrete quote checks

**Corollary 1.5** (Continuity and slope restrictions)**.** *European call and put prices are continuous in strike. Wherever the derivative exists,*

$$-D(t,T) \le c'(K) \le 0,$$
$$0 \le p'(K) \le D(t,T).$$

*For American options with $r \ge 0$, the corresponding bounds are $-1 \le C'(K) \le 0$ and $0 \le P'(K) \le 1$. Convexity implies that secant slopes increase from left to right.*

*Proof.* The Lipschitz bounds imply continuity directly: if $K_n \to K$, then $|c(K_n) - c(K)| \le D|K_n - K| \to 0$, and likewise for the other functions. Divide each strike-difference inequality by $K_2 - K_1 > 0$ and take a derivative limit where it exists.

For the secant statement, take $K_1 < K_2 < K_3$ and write

$$K_2 = \frac{K_3 - K_2}{K_3 - K_1} K_1 + \frac{K_2 - K_1}{K_3 - K_1} K_3.$$

For any convex price function $g$, convexity gives

$$(K_3 - K_1)g(K_2) \le (K_3 - K_2)g(K_1) + (K_2 - K_1)g(K_3),$$
$$(K_3 - K_2)[g(K_2) - g(K_1)] \le (K_2 - K_1)[g(K_3) - g(K_2)].$$

Divide by $(K_2 - K_1)(K_3 - K_2) > 0$ to obtain

$$\frac{g(K_2) - g(K_1)}{K_2 - K_1} \le \frac{g(K_3) - g(K_2)}{K_3 - K_2}.$$

□

*Remark* 1.26 (A practical mathematical check). For a finite European call grid, adjacent secant slopes must lie in $[-D, 0]$ and be nondecreasing. For puts they must lie in $[0, D]$ and be nondecreasing. These are useful necessary checks, alongside individual bounds and parity. They are not, by themselves, a complete test of consistency across all maturities and all traded instruments.

## 1.8 Dependence on maturity

*Source coverage: lecture slides 87–88..*

### 1.8.1 American options with more time

**Theorem 1.13** (American maturity monotonicity)**.** *For $t \le T_1 \le T_2$, with otherwise identical contract terms,*

$$C(K,t,T_2) \ge C(K,t,T_1),$$
$$P(K,t,T_2) \ge P(K,t,T_1).$$

*Proof.* Buy the longer-maturity option and sell the shorter-maturity option. Whenever the short option is exercised at $\tau \le T_1$, exercise the long option simultaneously. Its exercise date is permitted because $\tau \le T_1 \le T_2$, and the exercise receipts cancel the short obligation. If the short expires unexercised, the remaining long option has nonnegative value. Thus the position cannot have negative initial cost without arbitrage. This proves each inequality. □

*Remark* 1.27. The proof uses inclusion of exercise opportunities, not a comparison of the stock price at two different dates. It does not require $r \ge 0$.

### 1.8.2 Why European options need additional conditions

European options with different maturities pay at different dates and depend on different stock values. A later European expiry does not include the earlier exercise right.

**Example 1.11** (A longer European put can be cheaper)**.** Suppose $S_u = 0$ for all $u \ge t$, with $r > 0$ and $K > 0$. Then

$$\begin{aligned} p(K,t,T_1) &= Ke^{-r(T_1-t)}, \\ p(K,t,T_2) &= Ke^{-r(T_2-t)} \\ &= p(K,t,T_1)e^{-r(T_2-T_1)} \\ &< p(K,t,T_1) \qquad (T_2 > T_1). \end{aligned}$$

Both puts pay the same amount $K$, but the later payment is worth less now.

**Example 1.12** (A longer European call can be cheaper with dividends)**.** Let $r = 0$, $\delta > 0$, and suppose the ex-dividend stock evolves deterministically as $S_u = S_t e^{-\delta(u-t)}$. Reinvested total stock wealth is constant, consistent with the zero-rate bank account. Choose $T_1 < T_2$ with $S_{T_2} > K$. Then the deterministic European call values are

$$\begin{aligned} c(K,t,T_1) &= S_t e^{-\delta(T_1-t)} - K, \\ c(K,t,T_2) &= S_t e^{-\delta(T_2-t)} - K \\ &< c(K,t,T_1). \end{aligned}$$

Thus there is no unconditional maturity monotonicity for European calls when dividends are allowed.

**Corollary 1.6** (An important European-call exception)**.** *For a non-dividend-paying stock and* $r \geq 0$*,*

$$c(K, t, T_2) \geq c(K, t, T_1) \qquad (T_2 \geq T_1).$$

*Proof.* For each maturity, American and European call values coincide. Hence

$$\begin{aligned} c(K, t, T_2) &= C(K, t, T_2) \\ &\geq C(K, t, T_1) \\ &= c(K, t, T_1). \end{aligned}$$

□

*Remark* 1.28. The lecture's warning that European maturity comparisons need not hold is correct in general. It should not obscure this useful special case, which follows from results already proved in Part I.

## 1.9 Engineering payoffs with stocks, bonds, and options

*Source coverage: lecture slides 89–106..* For this section, all options are European with a common expiry $T$. Unless stated otherwise, assume no dividends, so a long share contributes terminal wealth $S_T$ without additional intermediate cash flows. With dividends, the formulas still describe the terminal ex-dividend positions, but total profit must also account for all distributions and financing. A prepaid share can be used when a terminal-only share claim is intended.

Write $s = S_T$. Every displayed payoff is a terminal value *before* subtracting financed setup cost. Negative initial cost means an initial credit, which must still be compared with future liabilities.

### 1.9.1 Protective puts: a floor on terminal value

**Definition 1.23** (Protective put or floor)**.** A protective put consists of one long share and one long put with strike $K$.

**Proposition 1.8** (Floor payoff and replication)**.** *The protective put has payoff*

$$s + (K - s)^+ = \max\{s, K\} = K + (s - K)^+.$$

*Its price is $S_t + p_t = KD + c_t$ in the no-dividend setting.*

*Proof.* If $s < K$, the put adds $K - s$, so the total is $K$. If $s \geq K$, the put pays zero, so the total is $s$. Also,

$$s + (K - s)^+ = K + (s - K)^+$$

by the positive-part identity. The right side is replicated by a $K$-face-value bond and a call. Equality of prices follows from no-arbitrage or put-call parity. □

If setup cost is $v = S_t + p_t$, financed profit satisfies

$$\Pi_T \geq K - ve^{r(T-t)}.$$

The floor is on terminal portfolio value, not a guarantee of nonnegative profit. Insurance has a cost.

### 1.9.2 Covered calls: selling the upper tail

**Definition 1.24** (Covered call)**.** A covered call consists of one long share and one short call on that share.

**Proposition 1.9** (Covered-call payoff)**.**

$$\begin{aligned} s - (s - K)^+ &= \min\{s, K\} \\ &= K - (K - s)^+. \end{aligned}$$

*Consequently its cost satisfies $S_t - c_t = KD - p_t$.*

*Proof.* For $s \leq K$, the call pays zero and the position is worth $s$. For $s > K$, the call liability is $s - K$, leaving $K$. The alternative expression follows from the positive-part identity, and prices follow by replication. □

*Remark* 1.29 (Premium is not guaranteed profit). Selling the call produces a known initial receipt, but the stock can still fall to zero. The payoff is capped above at $K$ and remains exposed to stock losses below $K$. Describing the premium as income does not remove the portfolio's risk or financing costs.

### 1.9.3 Protecting and covering short stock

**Definition 1.25** (Cap on a short position)**.** In the lecture's terminology, a cap consists of short stock plus a long call. Its terminal payoff is

$$-s + (s - K)^+ = -\min\{s, K\}.$$

For $s < K$, the payoff is $-s$; for $s \geq K$, it is $-K$. Thus the terminal cost of closing the short exposure is capped at $K$. Initial short-sale proceeds and the call premium must be included to obtain total profit.

**Definition 1.26** (Covered put)**.** A covered put consists of short stock plus a short put. Its terminal payoff is

$$-s - (K - s)^+ = -\max\{s, K\}.$$

For $s < K$, the payoff is $-K$; for $s \geq K$, it is $-s$. The premium compensates for giving up favourable outcomes from further stock-price declines, but unbounded loss from a large rise in the stock remains.

*Remark* 1.30 (Terminology). Here "floor" and "cap" refer to the stock-option combinations in the lecture. In other settings the same terms describe interest-rate derivatives; those are different contracts.

### 1.9.4 Bull and bear spreads with calls and puts

Fix $K_1 < K_2$ and $d = K_2 - K_1$.

**Definition 1.27** (Bull call spread)**.** Buy one call at $K_1$ and sell one call at $K_2$. The payoff is

$$H_{\mathrm{BC}}(s) = (s - K_1)^+ - (s - K_2)^+ = \begin{cases} 0, & s \leq K_1, \\ s - K_1, & K_1 < s < K_2, \\ d, & s \geq K_2. \end{cases}$$

This payoff increases with $s$ but is capped. Its cost is $v_{\mathrm{BC}} = c(K_1) - c(K_2)$, with $0 \leq v_{\mathrm{BC}} \leq dD$. Let $a = v_{\mathrm{BC}} e^{r(T-t)} \in [0, d]$. Then

$$\begin{aligned} \min_s \Pi_T &= -a, \\ \max_s \Pi_T &= d - a, \\ \text{interior break-even} &= K_1 + a. \end{aligned}$$

If $a = 0$ or $a = d$, a whole flat region may also have zero profit.

**Definition 1.28** (Bear put spread)**.** Buy one put at $K_2$ and sell one put at $K_1$. The payoff is

$$H_{\mathrm{BP}}(s) = (K_2 - s)^+ - (K_1 - s)^+ = \begin{cases} d, & s \le K_1, \\ K_2 - s, & K_1 < s < K_2, \\ 0, & s \ge K_2. \end{cases}$$

The cost is $v_{\mathrm{BP}} = p(K_2) - p(K_1)$, again between zero and $dD$. With $a = v_{\mathrm{BP}} e^{r(T-t)}$, the minimum and maximum financed profits are $-a$ and $d - a$, and the interior break-even is $K_2 - a$.

**The other two directional spreads**

A bear call spread sells the lower-strike call and buys the higher-strike call; its payoff is $-H_{\mathrm{BC}}$. A bull put spread buys the lower-strike put and sells the higher-strike put; its payoff is $-H_{\mathrm{BP}}$.

**Proposition 1.10** (Call and put spreads differ by a bond)**.** *For every $s$,*

$$\begin{aligned} H_{\mathrm{BC}}(s) + H_{\mathrm{BP}}(s) &= d, \\ H_{\mathrm{BC}}(s) - \left[(K_1 - s)^+ - (K_2 - s)^+\right] &= d. \end{aligned}$$

*Consequently,*

$$c(K_1) - c(K_2) + p(K_2) - p(K_1) = dD.$$

*Proof.* Subtract the positive-part identities at $K_1$ and $K_2$:

$$\begin{aligned} &(s - K_1)^+ - (K_1 - s)^+ \\ &\quad - \left[(s - K_2)^+ - (K_2 - s)^+\right] \\ &= (s - K_1) - (s - K_2) = K_2 - K_1 = d. \end{aligned}$$

Rearrange and apply the law of one price. Thus a bull call spread equals a bull put spread plus a bond paying $d$. The two have different initial costs but, after consistent financing, the same profit profile. □

### 1.9.5 Collars: imposing both a floor and a cap

**Definition 1.29** (Collar)**.** A collar consists of one long share, one long put at $K_L$, and one short call at $K_U$, where $K_L < K_U$.

**Proposition 1.11** (Collar payoff and decomposition)**.**

$$\begin{aligned} H_{\mathrm{col}}(s) &= s + (K_L - s)^+ - (s - K_U)^+ \\ &= \begin{cases} K_L, & s \le K_L, \\ s, & K_L < s < K_U, \\ K_U, & s \ge K_U \end{cases} \\ &= K_L + (s - K_L)^+ - (s - K_U)^+. \end{aligned}$$

*Its cost is*

$$S_t + p(K_L) - c(K_U) = K_L D + c(K_L) - c(K_U).$$

*Proof.* If $s \le K_L$, the put restores stock value to $K_L$, while the call is worthless. If $K_L < s < K_U$, both options are worthless. If $s \ge K_U$, the call removes all stock value above $K_U$. The last expression follows from $s + (K_L - s)^+ = K_L + (s - K_L)^+$. Its price is the cost of a bond paying $K_L$ plus a bull call spread. □

*Remark* 1.31 (Zero-cost collars). If $p(K_L) = c(K_U)$, the *option overlay* has zero initial cost. The stock still costs $S_t$. A zero-cost overlay is therefore not a zero-cost total investment.

*Remark* 1.32 (What no-arbitrage does and does not say). A collar can reduce downside exposure and change return variability. No-arbitrage alone does not prohibit an improvement in a chosen risk-adjusted performance statistic, nor does it specify expected returns. What it prohibits is a freely financed strategy with no losses and a genuine gain. The lecture's historical anecdote is not a mathematical proof about risk-adjusted performance.

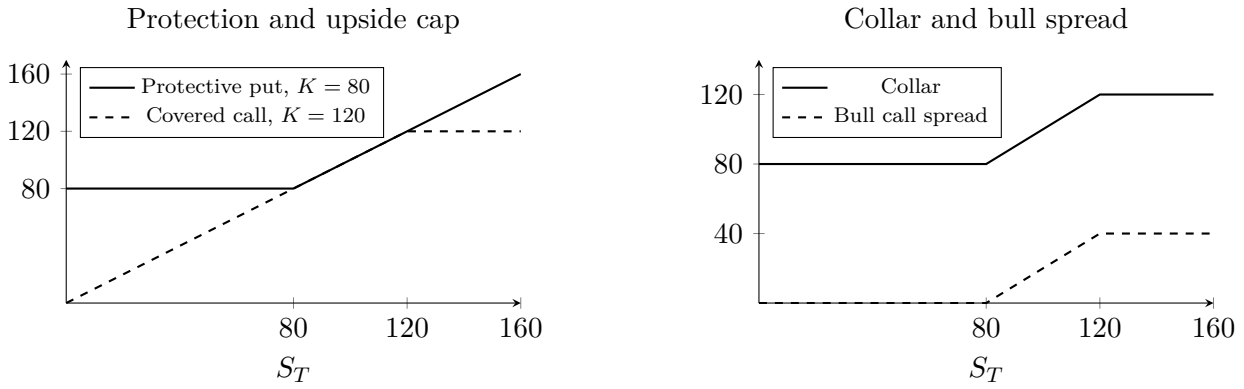



Figure 1.2: Exact terminal payoff profiles. The collar with strikes 80 and 120 equals the bull spread plus 80 dollars at maturity.

### 1.9.6 Straddles and strangles

**Definition 1.30** (Long straddle)**.** A long straddle buys a call and a put with the same strike $K$ and expiry. An at-the-money straddle chooses $K$ near the current spot price, but the payoff identity does not require that choice.

**Proposition 1.12** (Straddle payoff and break-even)**.** *The terminal payoff is*

$$\begin{aligned} H_{\text{strad}}(s) &= (s - K)^+ + (K - s)^+ \\ &= |s - K|. \end{aligned}$$

*If its premium is $v = c(K) + p(K)$ and $a = ve^{r(T-t)}$, its financed profit is $|s - K| - a$. Break-even values are $K - a$ and $K + a$ whenever they lie in the admissible domain $s \ge 0$.*

*Proof.* Use $x^+ + (-x)^+ = |x|$ with $x = s - K$. Solve $|s - K| = a$ to obtain the two break-even values. If $a > K$, the lower value is negative and cannot be attained by a nonnegative stock. □

The largest loss occurs at $s = K$ and equals the financed premium $a$. Profit requires a sufficiently large terminal deviation, not merely an intraday fluctuation that reverses before expiry.

**Definition 1.31** (Long strangle)**.** A long strangle buys a put at $K_L$ and a call at $K_U$, where $K_L < K_U$. A usual out-of-the-money strangle has $K_L < S_t < K_U$.

Its payoff is

$$\begin{aligned} H_{\text{strang}}(s) &= (K_L - s)^+ + (s - K_U)^+ \\ &= \begin{cases} K_L - s, & s < K_L, \\ 0, & K_L \le s \le K_U, \\ s - K_U, & s > K_U. \end{cases} \end{aligned}$$

Let $v = p(K_L) + c(K_U)$ and $a = ve^{r(T-t)}$. Break-even occurs at $K_L - a$ if nonnegative, and at $K_U + a$. The entire interval $[K_L, K_U]$ produces the maximum loss $a$.

*Remark* 1.33 (The meaning of "long volatility"). The straddle and strangle payoffs are convex functions of $s$. They benefit from sufficiently large terminal moves relative to their premiums. In a pricing model that relates greater dispersion to convex-payoff values, this motivates the phrase "long volatility". Part I has not yet introduced such a model: a payoff diagram alone does not establish a sensitivity to every volatility parameter or a response to every change in realized path variation.

### 1.9.7 Butterflies and the lecture's shifted convention

**Definition 1.32** (Standard symmetric long call butterfly)**.** Let $K_L < K_M < K_U$ with $K_M - K_L = K_U - K_M = h > 0$. Buy one call at $K_L$, sell two calls at $K_M$, and buy one call at $K_U$.

**Proposition 1.13** (Butterfly payoff)**.** *The standard butterfly payoff is*

$$\begin{aligned} B(s) &= (s - K_L)^+ - 2(s - K_M)^+ + (s - K_U)^+ \\ &= \begin{cases} 0, & s \le K_L, \\ s - K_L, & K_L < s \le K_M, \\ K_U - s, & K_M < s < K_U, \\ 0, & s \ge K_U. \end{cases} \end{aligned}$$

*In particular,* $0 \le B(s) \le h$. *Its cost* $b = c(K_L) - 2c(K_M) + c(K_U)$ *satisfies* $0 \le b \le hD$.

*Proof.* For $s \le K_L$, every call is worthless. For $K_L < s \le K_M$, only the first call contributes. For $K_M < s < K_U$,

$$\begin{aligned} B(s) &= (s - K_L) - 2(s - K_M) \\ &= 2K_M - K_L - s \\ &= K_U - s. \end{aligned}$$

For $s \ge K_U$,

$$\begin{aligned} B(s) &= s - K_L - 2s + 2K_M + s - K_U \\ &= 2K_M - K_L - K_U = 0. \end{aligned}$$

The peak at $K_M$ is $h$. Apply payoff bounds to obtain the cost bounds. Convexity independently gives $b \ge 0$. □

With $a = be^{r(T-t)} \in [0, h]$, financed profit is $B(s) - a$. Maximum loss is $a$, maximum profit is $h - a$, and the two interior break-even points are $K_L + a$ and $K_U - a$.

**Proposition 1.14** (Short straddle plus long strangle is a shifted butterfly)**.** *The lecture's combination is*

$$L(s) = (K_L - s)^+ + (s - K_U)^+ - (s - K_M)^+ - (K_M - s)^+.$$

*For symmetric strikes,*

$$L(s) = B(s) - h.$$

*Thus it has payoff $-h$ in both tails and zero at the middle strike. It is not the nonnegative standard butterfly payoff.*

*Proof.* Use $(K - s)^+ = (s - K)^+ + K - s$ twice:

$$\begin{aligned} L(s) &= (s - K_L)^+ + K_L - s + (s - K_U)^+ \\ &\quad - (s - K_M)^+ - \big((s - K_M)^+ + K_M - s\big) \\ &= (s - K_L)^+ - 2(s - K_M)^+ + (s - K_U)^+ + K_L - K_M \\ &= B(s) - h. \end{aligned}$$

Its initial value is consequently $\ell = b - hD$. After consistent financing,

$$\begin{aligned} L(s) - \ell e^{r(T-t)} &= B(s) - h - (b - hD)e^{r(T-t)} \\ &= B(s) - h - be^{r(T-t)} + h \\ &= B(s) - be^{r(T-t)}. \end{aligned}$$

The shifted construction and standard butterfly therefore have identical financed profit, despite different payoff levels and initial prices. □

**Example 1.13** (Strikes 80, 100, and 120)**.** Here $h = 20$. The standard butterfly pays zero outside $[80, 120]$ and 20 at $s = 100$. The lecture's short-straddle-plus-long-strangle combination pays $-20$ outside that interval and zero at $s = 100$. Adding a bond paying 20 converts the lecture's payoff into the standard one.

*Remark* 1.34 (Butterflies are not globally concave). The tent-shaped payoff rises and then falls, but its slope also jumps upward at the outer strikes. It is therefore not globally concave. A butterfly concentrates its favourable outcomes near the middle strike; describing it as "short volatility" is a qualitative description near that region, not a universal sign theorem for volatility sensitivity across all prices and models.

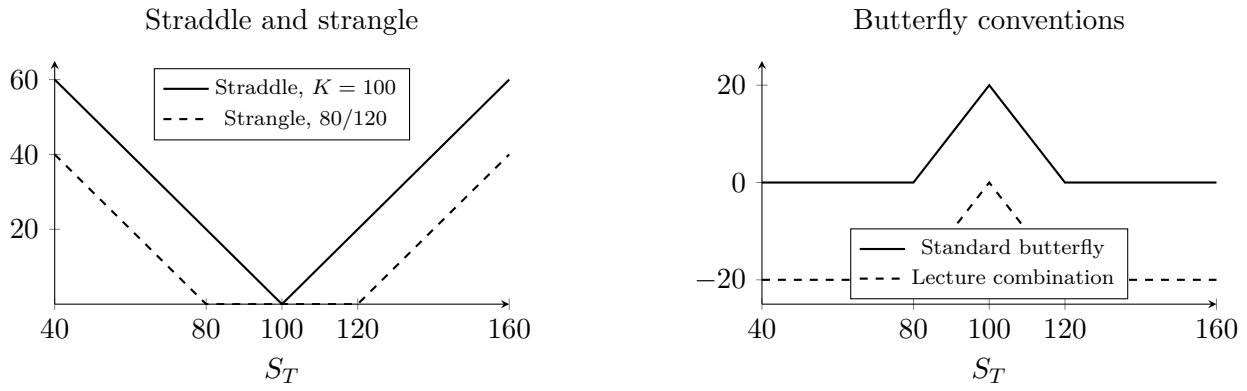



Figure 1.3: Exact payoff diagrams. The two butterfly conventions differ by a constant terminal payment; their consistently financed profits agree.

### 1.9.8 Reference table of terminal payoffs

| Strategy | Holdings | Terminal payoff |
|---|---|---|
| Synthetic long forward | Long call, short put at $K$ | $s - K$ |
| Protective put | Long share, long put | $\max\{s, K\}$ |
| Covered call | Long share, short call | $\min\{s, K\}$ |
| Cap on short stock | Short share, long call | $-\min\{s, K\}$ |
| Covered put | Short share, short put | $-\max\{s, K\}$ |
| Bull call spread | Long call $K_L$, short call $K_U$ | $(s - K_L)^+ - (s - K_U)^+$ |
| Bear put spread | Long put $K_U$, short put $K_L$ | $(K_U - s)^+ - (K_L - s)^+$ |
| Collar | Long share, put $K_L$, short call $K_U$ | $\min\{\max\{s, K_L\}, K_U\}$ |
| Straddle | Long call and put at $K$ | $\lvert s - K\rvert$ |
| Strangle | Long put $K_L$, long call $K_U$ | $(K_L - s)^+ + (s - K_U)^+$ |
| Symmetric butterfly | Calls at $K_L, K_U$, short two at $K_M$ | Tent payoff $B(s)$ |

### 1.9.9 Extension: replication of piecewise-linear payoffs

The strategies above are instances of one general construction. This is an explanatory extension beyond the lecture's individual examples.

**Theorem 1.14** (Call-basis representation)**.** *Let $h : [0, \infty) \to \mathbb{R}$ be continuous and piecewise linear, with breakpoints $0 < K_1 < \cdots < K_m$. Let its slopes on the consecutive intervals be $b_0, b_1, \ldots, b_m$, and let $a = h(0)$. Then*

$$h(s) = a + b_0 s + \sum_{j=1}^{m} (b_j - b_{j-1})(s - K_j)^+.$$

*If the required calls and prepaid share are traded, its time-$t$ price is*

$$aD + b_0 A + \sum_{j=1}^{m} (b_j - b_{j-1}) c(K_j, t, T).$$

*Proof.* Call the right side $g(s)$. At zero, $g(0) = a = h(0)$. On the first interval $[0, K_1)$, every positive-part term vanishes, so $g$ has slope $b_0$. On $(K_j, K_{j+1})$, exactly the first $j$ call terms have slope one, and therefore

$$\begin{aligned} g'(s) &= b_0 + \sum_{i=1}^{j} (b_i - b_{i-1}) \\ &= b_0 + (b_1 - b_0) + (b_2 - b_1) + \cdots + (b_j - b_{j-1}) \\ &= b_j. \end{aligned}$$

The same calculation works beyond the last breakpoint. Thus $g$ and $h$ have the same initial value, the same slope on every segment, and are continuous at the same breakpoints. They coincide everywhere. The formula describes a portfolio of a bond, prepaid stock, and calls, so the price follows from replication. □

*Remark* 1.35. A positive slope jump requires a long call; a negative slope jump requires a short call. This gives a systematic way to design a payoff rather than memorizing each named strategy.

## 1.10 What model-free pricing determines, and what it leaves open

*Source coverage: lecture slides 13, 55, 107–108; additional worked example below..*

### 1.10.1 Replicable versus nonreplicable payoffs

**Definition 1.33** (Replicable terminal claim)**.** A terminal claim $H$ is replicable if an admissible self-financing strategy has terminal wealth $H$ in every relevant state. Its no-arbitrage price equals the setup cost of that strategy.

A static portfolio of a bank account and a non-dividend stock pays $a + bS_T$, an affine function of the terminal stock price. A call pays $(S_T - K)^+$, which changes slope at $K$. If possible terminal stock values lie on both sides of $K$ in a sufficiently rich state space, a single affine function cannot generally match this kink.

*Remark* 1.36 (The scope of nonreplicability)*.* The lecture's statement that an option cannot be replicated by stock and bond is a statement about the general model-free setting. In a two-state one-period market, two independent traded payoffs can replicate every terminal claim. In some continuous-time models, dynamic stock-and-bond trading also replicates calls. These conclusions require structure not yet assumed in Part I.

### 1.10.2 A complete three-state example of price nonuniqueness

**Example 1.14** (Different arbitrage-free call prices with the same stock and bond)**.** Consider a one-period market with $r = 0$, $S_0 = 100$, and three possible stock values

$$S_T \in \{80, 100, 120\}.$$

Assume each state has positive physical probability. A call with strike 100 pays

$$H = (0, 0, 20)$$

across the three states. Suppose a static stock-and-cash portfolio replicated it. Then its coefficients $a, b$ would satisfy

$$\begin{aligned} a + 80b &= 0, \\ a + 100b &= 0, \\ a + 120b &= 20. \end{aligned}$$

Subtracting the first equation from the second gives $20b = 0$, so $b = 0$ and $a = 0$. The third equation would then read $0 = 20$, a contradiction. The call is not replicable with the existing assets.

For any $q \in (0, 1/2)$, define strictly positive state-price weights

$$(q_1, q_2, q_3) = (q, 1 - 2q, q).$$

These weights price the bond and stock correctly:

$$\begin{aligned} q_1 + q_2 + q_3 &= q + (1 - 2q) + q = 1, \\ 80q_1 + 100q_2 + 120q_3 &= 80q + 100(1 - 2q) + 120q \\ &= 100. \end{aligned}$$

Price every traded payoff by its weighted sum. The call then has price

$$c_0 = 0q + 0(1-2q) + 20q = 20q.$$

This pricing system is arbitrage-free. Indeed, if a portfolio has state payoffs $z_1, z_2, z_3 \geq 0$ and at least one is positive, its cost is

$$qz_1 + (1-2q)z_2 + qz_3 > 0,$$

so it cannot be a zero-cost arbitrage. If all payoffs are zero, its cost is zero, so it cannot produce a strictly positive initial receipt either.

For instance, $q = 0.2$ gives $c_0 = 4$ and $q = 0.4$ gives $c_0 = 8$. Both prices coexist with the same $S_0 = 100$, zero interest rate, and the same set of terminal stock values, in two different arbitrage-free price systems. No-arbitrage alone does not choose between them.

*Remark* 1.37 (State prices are not physical probabilities here). The weights were introduced as positive linear pricing coefficients. They happen to sum to one because $r = 0$. We have not asserted that they are the actual probabilities of the stock outcomes, or that physical expected payoff is the correct pricing rule.

### 1.10.3 A compact formula reference

Let $\Delta = T - t$, $D = e^{-r\Delta}$, and $A = S_t e^{-\delta\Delta}$ for a continuous dividend yield. Then

$$\begin{aligned} f_t(K,T) &= A - KD, & F(t,T) &= \frac{A}{D} = S_t e^{(r-\delta)\Delta}, \\ c_t - p_t &= A - KD, & c_t + KD &= p_t + A, \\ (A-KD)^+ \leq c_t &\leq A, & (KD-A)^+ \leq p_t &\leq KD. \end{aligned}$$

For known discrete dividends, replace $A$ by $S_t - \sum_i d_i e^{-r(t_i - t)}$.

If $r \geq 0$ and $\delta \geq 0$,

$$\begin{aligned} \max\{c_t, (S_t - K)^+\} \leq C_t \leq S_t, \\ \max\{p_t, (K - S_t)^+\} \leq P_t \leq K. \end{aligned}$$

If additionally $\delta = 0$,

$$\begin{aligned} C_t &= c_t, \\ S_t - K \leq C_t - P_t &\leq S_t - KD, \\ 0 \leq P_t - p_t &\leq K(1-D). \end{aligned}$$

For $K_1 < K_2$, European strike differences satisfy

$$\begin{aligned} 0 \leq c(K_1) - c(K_2) \leq D(K_2 - K_1), \\ 0 \leq p(K_2) - p(K_1) \leq D(K_2 - K_1). \end{aligned}$$

For American strike differences with $r \geq 0$, replace $D$ on the right by 1. All four strike-price functions are convex. American values are nondecreasing in maturity; European calls inherit this property when there are no dividends and $r \geq 0$.

### 1.10.4 Assumption checklist for proofs

| Result or construction | Essential qualification |
|---|---|
| Law of one price | Match all remaining cash flows, not only ex-dividend prices. |
| Discrete-dividend forward formula | Dividend amounts and dates are known, with delivery timing specified. |
| Continuous-yield prepaid stock | Dividends are reinvested; $e^{-\delta\Delta}$ is a share count. |
| American put bound $P_t \leq K$ | Nonnegative financing rate. |
| American call equals European call | No dividends and nonnegative financing rate. |
| European 1-Lipschitz strike bound | Nonnegative rate; otherwise use $D$-Lipschitz. |
| American convexity and strike bounds | A covering response at every short exercise time. |
| European maturity comparison | No unconditional ordering; use stated special cases. |
| Payoff-based trading profit | Subtract the financed initial cost and include intermediate cash flows. |
| Futures versus forwards | Settlement timing differs; quote equality needs financing assumptions. |

### 1.10.5 Where Part II begins

Part I determines exact prices for replicable linear claims and relations between option prices. It constrains many nonlinear claims without selecting a unique price. The natural next step is to specify a discrete-time stock model, identify attainable payoffs, and construct replicating strategies. In a one-period binomial model, there are two terminal stock states, so a stock-and-bond portfolio can solve two payoff equations. That is the starting point for the next part.

## 1.11 Worked diagnostic exercises

These exercises consolidate the constructions, including the proofs left open in the lecture. They introduce no additional pricing model.

### 1.11.1 A put quote below its lower bound

**Example 1.15.** Assume no dividends, $S_0 = 90$, $K = 100$, $r = 0$, and a European put quote $p_0 = 7$. The lower bound is $(100 - 90)^+ = 10$, so the put is underpriced.

Buy one put, buy one share, and borrow 100. Setup produces

$$-7 - 90 + 100 = 3.$$

The terminal payoff of those positions is

$$(100 - S_T)^+ + S_T - 100 = (S_T - 100)^+ \geq 0.$$

Keep the three-dollar receipt in the bank. The final wealth is at least three dollars in every state. This explicitly converts a lower-bound violation into arbitrage.

### 1.11.2 A vertical spread that is too expensive

**Example 1.16.** Let $K_1 = 100$, $K_2 = 110$, $r = 0$, and suppose European call quotes are $c(100) = 15$ and $c(110) = 3$. Their difference is 12, exceeding the maximum terminal spread payoff 10.

Sell the lower-strike call, buy the higher-strike call, and put 10 in the bank. The initial cash flow is

$$15 - 3 - 10 = 2.$$

The terminal payoff is

$$10 - (S_T - 100)^+ + (S_T - 110)^+ = 10 - H_{\mathrm{BC}}(S_T) \geq 0.$$

Banking the remaining two dollars yields guaranteed positive wealth. For general $r$, the bond needed for a European spread costs $10e^{-r(T-t)}$, whereas a hedge against American exercise at any time can use 10 dollars initially when $r \geq 0$.

### 1.11.3 Designing a capped loss-protection payoff

**Example 1.17.** Construct a terminal payoff that equals 90 below stock price 90, increases one-for-one from 90 to 110, and stays at 110 thereafter. This is

$$h(s) = \begin{cases} 90, & s \le 90, \\ s, & 90 < s < 110, \\ 110, & s \ge 110. \end{cases}$$

The initial slope is zero, the slope jumps up by one at 90 and down by one at 110. Hence the call-basis representation is

$$h(s) = 90 + (s - 90)^+ - (s - 110)^+.$$

Buy a bond paying 90, buy a call at 90, and sell a call at 110. Its price is

$$90D + c(90) - c(110).$$

Equivalently, buy prepaid stock, buy a put at 90, and sell a call at 110. Parity verifies the same cost:

$$A + p(90) - c(110) = 90D + c(90) - c(110).$$

### 1.11.4 Recovering a missing option quote

**Example 1.18.** Suppose $A = 98$, $KD = 100$, and a European call is priced at 6. Put-call parity requires

$$\begin{aligned} c - p &= A - KD = -2, \\ 6 - p &= -2, \\ p &= 8. \end{aligned}$$

The bound $p \ge KD - A = 2$ is satisfied, but it is parity with the traded call, rather than the bound alone, that identifies the put price. Without the call quote, many put prices could be consistent with the individual bounds.